Window Functions - Roman Urdu Cheat Sheet

> *Ye file tumhare liye hai taake window functions kabhi na bhoolo. Easy words me.*

1. Window Function hai kya?

Normal GROUP BY row ko khatam kar deta hai. Window function row ko khatam nahi karta, uske saath hi extra info jor deta hai.

Syntax:
`FUNCTION() OVER (PARTITION BY ... ORDER BY ... ROWS/RANGE ...)`

2. PARTITION BY - Dabba Banana

Matlab data ko groups me todna. Jaise category wise.

`PARTITION BY category_id` = Har category ka alag dabba.

Agar PARTITION BY nahi likhoge to poora table ek hi dabba ban jayega.

3. ORDER BY - Line Lagana

Dabbe ke andar tartib lagana. Kaun pehle, kaun baad me.

`RANK(), ROW_NUMBER(), LEAD(), LAG(), SUM()` jo running total hai, in sabke liye ORDER BY lazmi hai.

4. Main Functions

A) ROW_NUMBER() - Serial Number
Har row ko 1,2,3 number deta hai, chahe value same ho.

B) RANK() vs DENSE_RANK()
Dono rank dete hain lekin farq hai:
Data: 1000, 1000, 800

- RANK() = 1, 1, 3 (2 ko skip kar dega)
- DENSE_RANK() = 1, 1, 2 (skip nahi karega)
- ROW_NUMBER() = 1, 2, 3 (sabko alag)

C) SUM(), AVG(), COUNT() with OVER

`SUM(total) OVER (PARTITION BY category)` = Category ka total har row me likh dega.

D) LEAD() / LAG() - Aage Peeche dekhna

- `LAG(total) OVER (ORDER BY date)` = Pichli row ki value
- `LEAD(total) OVER (ORDER BY date)` = Agli row ki value

E) FIRST_VALUE() / LAST_VALUE()

- `FIRST_VALUE()` = Dabbe ki pehli value
- `LAST_VALUE()` = Dabbe ki aakhri value

LAST_VALUE ke liye ye frame lazmi likhna padta hai warna galat result dega:
`ROWS BETWEEN UNBOUNDED PRECEDING AND UNBOUNDED FOLLOWING`

5. ROWS vs RANGE - Sabse Confusing

ROWS = Physical Row ginta hai
Har row alag, chahe value same ho.

RANGE = Value dekhta hai
Jinki ORDER BY wali value same hai, unko ek hi samajhta hai.

Example: Ali 1000, Sara 1000, Ahmed 800

- ROWS wala SUM: Ali=1000, Sara=2000, Ahmed=2800
- RANGE wala SUM: Ali=2000, Sara=2000, Ahmed=2800 (1000 walo ko jor diya)

> *99% time ROWS use karo taake confusion na ho.*
> *Agar value unique hai to dono ka result same hota hai. Duplicate par farq aata hai.*

6. Frame - Kitna Hissa Lena Hai?

Formula: `ROWS / RANGE BETWEEN [Shuru] AND [Khatam]`

Shuru/Khatam par ye 5 cheezein aa sakti hain:

1. `UNBOUNDED PRECEDING` = Bilkul shuru se (Be-had)
2. `n PRECEDING` = Mujhse n pehle se (Had wala) -> Shamil karta hai, chorta nahi
3. `CURRENT ROW` = Jahan main khada hu
4. `n FOLLOWING` = Mujhse n baad tak (Had wala)
5. `UNBOUNDED FOLLOWING` = Bilkul aakhir tak (Be-had)

Sabse zyada use hone wale 4 Frames:

1. Full Window:
`ROWS BETWEEN UNBOUNDED PRECEDING AND UNBOUNDED FOLLOWING` -> Poora dabba

2. Running Total (Default):
`ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW` -> Shuru se ab tak

3. Moving Average (3 din ka):
`ROWS BETWEEN 2 PRECEDING AND CURRENT ROW` -> Main + 2 pichle

4. Sliding Window (3 rows):
`ROWS BETWEEN 1 PRECEDING AND 1 FOLLOWING` -> 1 peeche + Main + 1 aage

7. BOUNDED vs UNBOUNDED

- BOUNDED = Had wala (1 PRECEDING, CURRENT ROW)
- UNBOUNDED = Be-had (UNBOUNDED PRECEDING, UNBOUNDED FOLLOWING)

- UN ka matlab ulta = Be-had.

8. Sabse Important Rules

Rule 1: WHERE me Window Function nahi chalega
Galat: `WHERE RANK() OVER(...) = 1`
Sahi: CTE banao phir filter karo

```
1  WITH Ranked AS (SELECT *, RANK() OVER(...) as rnk FROM table)
1  SELECT * FROM Ranked WHERE rnk = 1
```

Rule 2: CURRENT ROW kahan se pata chalta hai?
SQL ORDER BY se line lagata hai aur har row par ja kar bolta hai "ab CURRENT ROW ye hai". Is liye ORDER BY lazmi hai.

Rule 3: ROWS likhna zaroori hai?
Agar BETWEEN likh rahe ho to ROWS ya RANGE likhna lazmi hai. Agar frame likho hi nahi to SQL default `RANGE BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW` laga deta hai.

Final Yaad Karne ka Tarika

PARTITION BY = Dabba banao
ORDER BY = Line lagao
ROWS vs RANGE = Row gino ya Value dekho
CURRENT ROW = Jahan tum khade ho